

# SHUBHAM MATHUR

## Senior Software Engineer

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### SUMMARY

Senior Software Engineer with 8+ years across the **full stack** (web, backend, cloud/DevOps) and deep backend and distributed-systems expertise in **identity/auth, security, and production GenAI**. Architected compliance and auth platforms for **4.5M+ users** and shipped an enterprise **RAG** assistant end-to-end. Work AI-natively with **Claude Code**: agentic, plan-driven development (subagent orchestration, MCP tooling) from design through implementation and review. Open-source contributor at **Stripe, Apple, and Wise**.

### EXPERIENCE

#### Staff Software Engineer

Apr 2026 – Present

##### CivilGrid

Remote, India

- **Technical leadership:** Establishing CivilGrid's India engineering presence through hiring, engineering and testing standards, code-review culture, and architectural direction, while delivering full-stack features end-to-end across a GIS web platform (React, TypeScript, PostgreSQL).
- **Architecture ownership:** Led migration of a static configuration system (214 configs) to a database-driven, feature-flagged, Redis-cached architecture with parity-checking, owning the dependency mapping and data-access contract.

#### Senior Software Engineer

Jan 2024 – Jan 2026

##### Vimeo

Remote, India

- **Regulatory compliance (Age Assurance):** Technical lead and architect, authored the technical specification and designed Vimeo's Zero-PII age-verification platform (5 platforms, **4.5M+ users**), shielding the company from UK OSA / EU DSA fines of up to **10% of global revenue**; built the analytics that drove C-level rollout decisions, accelerating compliance by around 20 days.
- **Identity modernization:** Designed and built a Go auth microservice (REST + gRPC) to carve auth out of a 15-year PHP monolith via a Strangler migration mapping **40+ integration points**; found and fixed a critical MFA-bypass plus XSS/open-redirect flaws.
- **Trust & Safety:** Sole technical owner, re-architected a broken user- and clip-deactivation pipeline (**195K+** "zombie" accounts still able to log in; GDPR risk) into a DDD + RabbitMQ system that decouples account and content deletion: cut processing from days to **under 2 minutes**, scaled to 50K+ accounts/batch and 1.6M+ clips/account, reconciling 1M+ orphaned records.
- **Applied GenAI:** Built and shipped VimeoIQ, an internal **RAG assistant** (GCP, Vertex AI Gemini, Vector Search) answering employee questions over Google Docs for India and US engineering teams, with source citations, confidence scoring, and pre-LLM PII scrubbing.

#### Engineer 2

Nov 2021 – Jan 2024

##### McKinsey & Company

Bengaluru, India

- **Early GenAI (banking IVR):** Designed and built a full-stack conversational GenAI application for enterprise banking IVR (NestJS, OpenAI; web, backend, and DevOps): LLM integration with PII redaction, stateful multi-turn orchestration (Redis), intent detection, and prompt-injection guardrails for real-time voice.
- **Healthcare platform:** Built the reliable SMS reminder system (event-driven Azure Functions + Service Bus + Twilio, fault-tolerant, multi-timezone scheduling) plus frontend business flows and login/signup for a HIPAA-compliant medication-adherence app, which contributed to a **34% drop in hospital readmissions**.
- **Scale & leadership:** Cut API latency **50%** on services for **500K+ users**; led architecture reviews, mentored 4 engineers, and set API and CI/CD standards.

#### Engineer 1

Jul 2018 – Nov 2021

##### McKinsey & Company

Bengaluru, India

- **Multi-client scale:** Delivered platforms for banking, insurance, and telecom clients across Malaysia, Singapore, Thailand, and the US (up to **20M+ users**); advanced from intern to Engineer 1.
- **Identity & distributed systems:** Built RBAC/ACLs and Okta SSO for a consultant engagement platform (GraphQL/PostgreSQL, MVP to org-wide); built REST microservices over legacy government-agency SOAP integrations for a Malaysian bank (+60% RM productivity).

### OPEN SOURCE

**Merged PRs:** [Stripe / link-cli](#) · [Apple / password-manager-resources](#) · [Wise / PipelineWise](#) · [Reactive Resume](#)

### SKILLS

**Languages:** JavaScript, TypeScript, SQL | *Familiar:* Go, PHP, Python, Java

**Backend:** Node.js, NestJS, Express.js | *Familiar:* Spring Boot, FastAPI

**Frontend:** React, Next.js, Tailwind, Jest, Web Security (CSP/XSS)

**Data:** PostgreSQL, MySQL, Redis, MongoDB | *Familiar:* Snowflake

**Messaging:** RabbitMQ, Azure Service Bus | *Familiar:* Kafka

**Cloud & DevOps:** Docker, GitHub Actions, GCP (Cloud Run, Vertex AI) | *Familiar:* AWS, Azure, Kubernetes

**AI / GenAI:** RAG, LLM orchestration, prompt engineering, embeddings & vector search, function calling, PII & prompt-injection guardrails

**AI Dev Tooling:** Claude Code, Cursor (agentic, plan-driven development; subagent orchestration, MCP tooling)

**Architecture:** Microservices, DDD, Event-Driven, gRPC, REST/GraphQL, OAuth2/JWT, Observability

### EDUCATION

**Vellore Institute of Technology**  
**B.Tech, Information Technology**

2018  
Vellore, India